
Conestoga College
MATH73235: Advanced Topics in Mathematics
Spring 2025 Midterm 2
July 15, 2025

Exam Duration: 5pm - 7pm.

Student Name (print): JOSHUA DIPLOMA

Student ID: 7897960

This exam contains 10 pages (including this cover page) and 5 questions. The total number of possible points is 50. Enter your answers in the space provided.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided.
- **Mysterious or unsupported answers will not receive full credit.** Correct answers should be supported by calculations, explanations, or algebraic work to receive full credit; incorrect answers supported by substantially correct calculations and explanations may still receive partial credit.
- **Provide exact answers** unless otherwise instructed.
- **Simplify all answers as much as possible.** This means that you need to combine like terms, reduce fractions, etc.
- **Be sure to state units for applied problems.**
- **Clearly identify your answer for each problem.**

Question	Points	Score
1	10	7
2	10	10
3	10	10
4	10	7
5	10	10
Total:	50	44

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1. (10 points) **Conceptual Questions.** Write an equation, an expression, or a short sentence to describe each of these concepts.

- (a) The expression for the volume integral $\iiint_T f(x, y, z) dx dy dz$ when the coordinates are changed from (x, y, z) to some other coordinates (u, v, w) .

Solution.

$$2 \quad \iiint f(x(u, v, w), y(u, v, w), z(u, v, w)) \frac{\partial(x, y, z)}{\partial(u, v, w)} du dv dw$$

- (b) The main equation for **Stokes' Theorem** for a vector field $\vec{F}$ in a surface S with a boundary curve C traversed such that S remains on the left of C .

Solution.

$$0 \quad \oint_C (\vec{\nabla} \cdot \vec{F}) \cdot \hat{n} ds \quad X \quad \text{Stokes:} \quad \iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} dA = \oint \vec{F} \cdot d\vec{r}$$

- (c) An expression for a vector $\vec{N}$ that is normal to the surface $z - x^2 - y^2 = 0$.

Solution.

$$2 \quad \vec{N} = \vec{\nabla} f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] = [-2x, -2y, 1]$$

- (d) The vector field $\vec{F}$ polar coordinates written as $F_r \hat{e}_r + F_\theta \hat{e}_\theta$ if in Cartesian coordinates we have $\vec{F} = F_1 \hat{i} + F_2 \hat{j} = (x^2 + y^2) \hat{i} - 2y \hat{j}$. Hint: $F_r = F_1 \cos \theta + F_2 \sin \theta$, $F_\theta = -F_1 \sin \theta + F_2 \cos \theta$.

Solution.

$$1 \quad x = r \cos \theta, \quad y = r \sin \theta, \quad x^2 + y^2 = r^2$$

$$\vec{F} = r^2 \hat{i} - 2r \sin \theta \hat{j} \rightarrow F_r \hat{e}_r + F_\theta \hat{e}_\theta?$$

$$F_r = ?$$

$$F_\theta = ?$$

- (e) The expression for the divergence $\nabla \cdot \vec{F}$ in spherical coordinates (ρ, θ, ϕ) if we are given that the divergence of the same vector field is $\nabla \cdot \vec{F} = r^2 + z^2$ in cylindrical coordinates (r, θ, ϕ) .

Solution.

$$2 \quad \vec{\nabla} \cdot \vec{F} = r^2 + z^2 \rightarrow \text{cylindrical}$$

$$\vec{\nabla} \cdot \vec{F} = \rho^2 \rightarrow \text{spherical}$$

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2. (10 points) Use the appropriate Lamé coefficients to compute the Laplacian of the scalar function

$$f(x, y, z) = x^2 + y^2 + z^2$$

in both Cylindrical and Spherical coordinates. You must compute all the required partial derivatives.

Solution.

Cylindrical: $h_1 = h_r = 1$, $h_2 = h_\theta = r$, $h_3 = h_z = 1$

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial r} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial z} \right) \right]$$

$$\left. \begin{array}{l} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{array} \right\} f(x, y, z) = x^2 + y^2 + z^2 \Rightarrow f(r, \theta, z) = r^2 \cos^2 \theta + r^2 \sin^2 \theta + z^2$$

$$= r^2 (\cos^2 \theta + \sin^2 \theta) + z^2 = r^2 + z^2$$

$$\frac{\partial f}{\partial r} = 2r, \quad \frac{\partial f}{\partial \theta} = 0, \quad \frac{\partial f}{\partial z} = 2z$$

$$\nabla^2 f = \frac{1}{r} \left[\frac{\partial}{\partial r} \left(r (2r) \right) + \frac{\partial}{\partial \theta} \left(\frac{1}{r} (0) \right) + \frac{\partial}{\partial z} \left(r (2z) \right) \right]$$

$$\nabla^2 f = \frac{1}{r} \left(\frac{\partial}{\partial r} (2r^2) \right) + \frac{1}{r} \frac{\partial}{\partial z} (2rz)$$

$$\nabla^2 f = \frac{1}{r} (4r) + 2$$

$$\boxed{\nabla^2 f = 6}$$



see last page

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3. (10 points) Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

Solution.

$$\phi = \text{const} \quad dA = \rho^2 \sin \phi \, d\rho \, d\theta$$

$$\int dA = \int \rho^2 \sin \phi \, d\rho \, d\theta$$

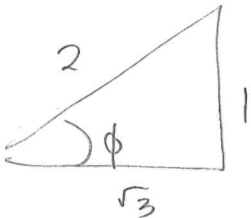
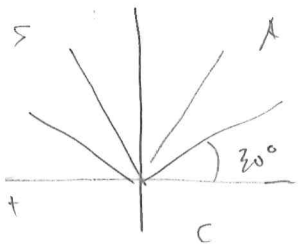
$$A = \sin \phi \int_0^2 \rho^2 \, d\rho \int_0^{2\pi} d\theta \quad ; \quad \phi = \frac{\pi}{6}$$

$$A = \sin\left(\frac{\pi}{6}\right) \left[\frac{\rho^3}{3} \right]_0^2 \left[\theta \right]_0^{2\pi}$$

$$A = \left(\frac{1}{2}\right) \left(\frac{8}{3}\right) (2\pi)$$

$$A = \frac{8}{3} \pi$$

$$\boxed{A = 2\pi} \quad \checkmark$$



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4. (10 points) Verify Gauss' Divergence Theorem by computing the following surface integral:

$$\iint_S \vec{F} \cdot \hat{n} \, dA,$$

and the volume integral:

$$\iiint_T (\nabla \cdot \vec{F}) \, dV,$$

where $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$, S is the surface of a sphere of radius 2 that forms the boundary of T .

Hint: This problem may be easier to solve in the spherical coordinates. Note that in spherical coordinates, $\vec{F} = \rho\hat{e}_\rho$, while for S , $\hat{n} = \hat{e}_\rho$, and thus

$$\vec{F} \cdot \hat{n} = (\rho\hat{e}_\rho) \cdot (\hat{e}_\rho) = \rho.$$

Solution.

Vol. integral

$$\vec{\nabla} \cdot \vec{F} = 1 + 1 + 1 = 3$$

$$\iiint (\nabla \cdot \vec{F}) \, dV \rightarrow \iiint 3 \, dx \, dy \, dz$$

$$= 3 \iiint \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$$

$$= 3 \int_0^2 \rho^2 \, d\rho \int_0^{2\pi} d\theta \int_0^\pi \sin \phi \, d\phi$$

$$= 3 \left[\frac{\rho^3}{3} \right]_0^2 \left[\theta \right]_0^{2\pi} \left[-\cos \phi \right]_0^\pi$$

$$= 3 \left(\frac{8}{3} \right) (2\pi) \left(-\cos(\pi) + \cos(0) \right)$$

$$\rightarrow = 32\pi$$



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$$F = x\hat{i} + y\hat{j} + z\hat{k}$$

Continue problem 4

Solution.

$$\oint F \cdot \hat{n} \, dA \quad F = [p \cos \theta \sin \phi, p \sin \theta \sin \phi, p \cos \phi]$$

④ continued ...

$$\vec{F} \cdot \hat{n} = p \hat{e}_r \cdot \hat{e}_r = p$$

$$\therefore \vec{F} \cdot \hat{n} = p \cos \theta \sin \phi + p \sin \theta \sin \phi + p \cos \phi \quad \times$$

$$\iint \vec{F} \cdot \hat{n} \, dA = \iint p \cos \theta \sin \phi + p \sin \theta \sin \phi + p \cos \phi \, d\theta \, d\phi, \quad p = \text{const}$$

$$= p \iint (\cos \theta \sin \phi + \sin \theta \sin \phi + \cos \phi) \, d\theta \, d\phi$$

$$= p \iint \cos \phi + \sin \phi (\cos \theta + \sin \theta) \, d\theta \, d\phi$$

$$= p \int_0^\pi \int_0^{2\pi} \cos \phi + \sin \phi (\cos \theta + \sin \theta) \, d\theta \, d\phi$$

$$= p \int_0^\pi \left[\theta \cos \phi + \sin \phi \left(\overbrace{\sin \theta}^{=0} - \overbrace{\cos \theta}^{=1} \right) \right] \Big|_0^{2\pi} \, d\phi$$

$$= p \int_0^\pi (2\pi) \cos \phi \, d\phi$$

$$= p \left[2\pi \sin \phi \right] \Big|_0^\pi$$

$$= 0$$

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5. (10 points) In the first quadrant of the Cartesian plane (i.e., $x \geq 0, y \geq 0$), we may define a set of coordinates known as the hyperbolic coordinates (u, v) , with transformation function

$$x = ve^u, \quad y = ve^{-u}.$$

Find the area of the triangular region R given by

$$0 \leq v \leq u, \quad 0 \leq u \leq 3.$$

Solution.

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$\left\{ \begin{array}{l} \frac{\partial x}{\partial u} = \frac{\partial}{\partial u}(ve^u) = (1)(ve^u) = ve^u \\ \frac{\partial x}{\partial v} = \frac{\partial}{\partial v}(ve^u) = (1)e^u = e^u \\ \frac{\partial y}{\partial u} = \frac{\partial}{\partial u}(ve^{-u}) = (-1)(ve^{-u}) = -ve^{-u} \\ \frac{\partial y}{\partial v} = \frac{\partial}{\partial v}(ve^{-u}) = (1)(e^{-u}) = e^{-u} \end{array} \right\} \begin{vmatrix} ve^u & e^u \\ -ve^{-u} & e^{-u} \end{vmatrix}$$

$$h_u = \sqrt{(ve^u)^2 + (-ve^{-u})^2}$$

$$h_v = \sqrt{(e^u)^2 + (e^{-u})^2}$$

$$h_u = \sqrt{v^2 e^{2u} + v^2 e^{-2u}}$$

$$h_v = \sqrt{e^{2u} + e^{-2u}}$$

$$h_u = \sqrt{v^2 (e^{2u} + e^{-2u})}$$

$$h_v = e^{\frac{2u}{2}} + e^{-\frac{2u}{2}}$$

$$h_u = v \left[e^{\frac{2u}{2}} + e^{-\frac{2u}{2}} \right]$$

$$h_v = e^u + e^{-u}$$

$$h_u = v [e^u + e^{-u}]$$

$$\int dA = \int h_u h_v du dv$$

$$\frac{\partial(x,y)}{\partial(u,v)} = ve^u e^{-u} - (-ve^{-u})(e^u)$$

$$= ve^{u-u} + ve^{-u+u}$$

$$\frac{\partial(x,y)}{\partial(u,v)} = 2v \quad \checkmark$$

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$$\int u \, dv = uv - \int v \, du$$

$$u = u$$

$$du = du$$

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⑤ continue

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$$\int dA = \int \ln uv \cdot \frac{\partial(x,y)}{\partial(u,v)} \, du \, dv$$

$$A = \iint v [e^u + e^{-u}] [e^u + e^{-u}] \, 2v \, du \, dv$$

$$2v^2 [e^u e^u + \overset{=1}{e^u e^{-u}} + \overset{=1}{e^{-u} e^u} + e^{-u} e^{-u}]$$

$$A = \iint 2v^2 [e^{2u} + e^{-2u} + 2] \, dv \, du$$

$$A = \int_0^3 \int_0^u 2v^2 [e^{2u} - e^{-2u} + 2] \, dv \, du$$

$$A = \int_0^3 \left[\frac{2v^3}{3} [e^{2u} - e^{-2u} + 2] \right] \bigg|_0^u \, du$$

$$A = \int_0^3 \left(\frac{2}{3} u^3 [e^{2u} - e^{-2u} + 2] \right) \, du$$

$$0 \leq v \leq u$$

$$0 \leq u \leq 3$$

$$\text{let } x = 2u$$

$$dx = 2du$$

$$\int e^x \, dx$$

$$du = \frac{1}{2} dx$$

$$\frac{1}{2} e^x \, dx$$

OK

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② continue...

Extra Page: Must submit even if unused.

SPHERICAL: $h_1 = h_\rho = 1$ $h_2 = h_\theta = \rho \sin \phi$ $h_3 = h_\phi = \rho$

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial \rho} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial \rho} \right) + \frac{\partial}{\partial \theta} \left(\frac{h_1 h_3}{h_2} \frac{\partial f}{\partial \theta} \right) + \frac{\partial}{\partial \phi} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial \phi} \right) \right]$$

$$\begin{aligned} x &= \rho \cos \theta \sin \phi \\ y &= \rho \sin \theta \sin \phi \\ z &= \rho \cos \phi \end{aligned} \quad \left. \begin{aligned} f(\rho, \theta, \phi) &= \rho^2 \cos^2 \theta \sin^2 \phi + \rho^2 \sin^2 \theta \cos^2 \phi + \rho^2 \sin \phi \\ &= \rho^2 \left[\cancel{\sin^2 \phi} + \cos^2 \phi \left(\cancel{\cos^2 \theta} + \cancel{\sin^2 \theta} \right) \right] \\ &= \rho^2 \end{aligned} \right\}$$

$$\frac{\partial f}{\partial \rho} = 2\rho \quad \frac{\partial f}{\partial \theta} = 0 \quad \frac{\partial f}{\partial \phi} = 0$$

$$\nabla^2 f = \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} \left((\rho^2 \sin \phi) (2\rho) \right) \right]$$

$$\nabla^2 f = \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} (2\rho^3 \sin \phi) \right]$$

$$\nabla^2 f = \frac{1}{\rho^2 \sin \phi} [6\rho^2 \sin \phi]$$

$$\boxed{\nabla^2 f = 6}$$

Formula Sheet

General Coordinates:

$$(x_1, x_2, x_3) \rightarrow (q_1, q_2, q_3):$$

$$x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3)$$

$$\text{Derivative Matrix: } D = \begin{bmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_1}{\partial q_2} & \frac{\partial x_1}{\partial q_3} \\ \frac{\partial x_2}{\partial q_1} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_2}{\partial q_3} \\ \frac{\partial x_3}{\partial q_1} & \frac{\partial x_3}{\partial q_2} & \frac{\partial x_3}{\partial q_3} \end{bmatrix}$$

$$\text{Lamé coefficients: } h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2$$

$$\text{Jacobian: } \frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)} = |J| = |\det D| = |h_1 h_2 h_3|$$

$$\text{Line element: } ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2$$

$$\text{Volume element: } dV = h_1 h_2 h_3 dq_1 dq_2 dq_3$$

$$\text{Area element for surface with } q_j = \text{const} \ (i, j, k \text{ distinct}): dA = h_i h_k dq_i dq_k$$

$$\text{Gradient for scalar function } f: \nabla f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} \hat{e}_{q_1} + \frac{1}{h_2} \frac{\partial f}{\partial q_2} \hat{e}_{q_2} + \frac{1}{h_3} \frac{\partial f}{\partial q_3} \hat{e}_{q_3}$$

$$\text{Divergence and curl of vector field } \vec{F} = F_{q_1} \hat{e}_{q_1} + F_{q_2} \hat{e}_{q_2} + F_{q_3} \hat{e}_{q_3}:$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 F_{q_1}) + \frac{\partial}{\partial q_2} (h_3 h_1 F_{q_2}) + \frac{\partial}{\partial q_3} (h_1 h_2 F_{q_3}) \right]$$

$$\nabla \times \vec{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_{q_1} & h_2 \hat{e}_{q_2} & h_3 \hat{e}_{q_3} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_{q_1} & h_2 F_{q_2} & h_3 F_{q_3} \end{vmatrix}$$

Laplacian of scalar field f :

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right].$$

Cylindrical Coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

Spherical Coordinates:

$$x = \rho \cos \theta \sin \phi, \quad y = \rho \sin \theta \sin \phi, \quad z = \rho \cos \phi$$

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$